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Runway scheduling during winter operations☆

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ABSTRACT

This paper presents an optimization model for the runway scheduling problem under consideration of winter operations. During periods of snowfall, runways have to be intermittently closed in order to clear them from snow, ice, and slush. To support human planners with the resulting complex scheduling tasks, we propose an integrated optimization model to simultaneously plan snow removal for multiple runways and to assign runways as well as take-off and landing times to aircraft. We formulate the model as a mixed-integer linear problem. To improve the computational tractability of our exact approach, we develop pruning rules and valid inequalities. Additionally, we derive initial start solutions heuristically. We validate and benchmark the model with realistic data from a large international airport and compare the results to a practice-based heuristic approach. We also demonstrate the applicability of our algorithm to large-scale aircraft landing instances from the literature. A computational study shows that our solution approach computes runway schedules which cause significantly less aircraft delay cost within a few seconds.

1. Introduction

Runway availability is a scarce resource at many major international airports. During snowfall, the availability of runways is further limited. Heavy snowfalls can require airport operations to be completely shut down. During light or medium snowfall, however, runways are usually intermittently closed in order to clear them from snow, ice, and slush, while airport operations continue. During peak periods, large airports operate at utilization levels at which the complete closure of a runway for snow removal temporarily decreases the runway capacity below demand for arriving and departing aircraft. In these situations, aircraft are often delayed due to limited runway capacity, which causes additional cost and effort for airports, airlines, and passengers. Moreover, aircraft delay may occasionally propagate throughout the network, potentially causing disruptions and cost at other airports not necessarily affected by snow [1]. While the frequency of such weather events is different for each airport and each year, airports in cities like Chicago, Frankfurt, Munich, or Prague typically prepare for winter operations from November to April.

In this paper, we show that an integrated scheduling of aircraft and snow removals on runways can reduce aircraft delay and associated cost. We solve the operational problem of assigning runways as well as take-off and landing times to aircraft while simultaneously planning snow removal activities on runways, considering the scarcity of snow removal equipment. This scarcity results from the condition that the number of snow removal groups available at airports is usually smaller than the number of runways which are operated in parallel. Such circumstances require a scheduling of snow removals which is synchronized with runway operations [1–3]. This results in a complex decision problem since a human planner, i.e., the responsible air traffic controller or runway manager, has to make fast decisions taking into account frequently changing conditions, e.g.,

- the set of aircraft to be considered often changes since aircraft are constantly taking-off or landing and new aircraft are entering the near-terminal airspace,
- attributes of aircraft such as the target take-off or landing time and the cost factor for delay are aircraft specific and can change over time, and
- the planning of snow removal activities is highly dependent on the current weather and its forecast as well as on the availability of snow removal equipment.

To account for changing and evolving decision variables and parameters, especially modifications in the set of considered aircraft, a recalculation of the optimal solution is necessary approximately once per minute. Due to the large number of considered aircraft, runways, and snow removal groups as well as the incorporation of

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the time dimension, the possible solution space is extremely large, which further complicates the planning task. Additionally, the multitude of involved stakeholders, such as passengers, airlines, air traffic control, and the airport operator, require a fair, transparent, and traceable decision. In practice, air traffic controllers and runway managers conduct the aircraft scheduling process manually based on a First-Come-First-Served (FCFS) principle, i.e., aircraft are sequenced according to their target take-off or landing time on the first available runway. The decision when and in which order runways are closed for snow removal is based on human experience and intuition instead of data-driven mathematical optimization. To facilitate these planning activities for aircraft and snow removals, we propose an optimization based solution methodology for the underlying scheduling tasks.

This paper contributes to the field of airport operations by presenting the first integrated model which simultaneously schedules departing and arriving aircraft and snow removals on runways. Integrating these planning decisions minimizes aircraft delay and associated cost, mostly, by reducing the duration of runway closures and by scheduling snow removals intelligently in between aircraft movements. So far, the runway scheduling problem during winter operations has not been studied in the literature. We model the problem as a static and deterministic mixed-integer linear problem. To take into account that considered aircraft and problem parameters change over time, we propose to implement the decisions for the first minute of the schedule and to recompute optimal schedules every minute with a planning horizon shifted by one minute. To achieve the required short computational times, we derive pruning rules to presolve and fix binary variables during preprocessing and we prove problem-specific valid inequalities which exploit compulsory precedence relations between aircraft and snow removals. Our proposed methodology enables planners to solve instances of realistic size to optimality. We show that our optimal approach significantly outperforms a practice-based heuristic in terms of overall delay cost.

The structure of this paper is as follows. In Section 2, we detail the runway scheduling problem during winter operations. Section 3 discusses previous work and related research articles and positions our work within the existing literature. In Section 4, we present a solution methodology for the considered scheduling problem, including a mathematical mixed-integer linear model formulation, pruning rules, valid inequalities, and a heuristic to derive initial start solutions. In Section 5, we apply our model to realistic instances of a large international airport and to large-scale instances from the literature. We compare the results to a practice-based heuristic mimicking the manual process performed by runway managers and air traffic controllers and we provide managerial remarks. Finally, we summarize and conclude our work in Section 6.

2. Problem description

For operational runway scheduling at airports, air traffic controllers assign runways and take-off and landing times to departing and arriving aircraft. This classical aircraft scheduling problem usually optimizes an operational planning horizon of up to two hours covering the time span between the next 16 to 120 minutes. In the final 15 minutes before take-off or landing, runway schedules can usually not be changed anymore due to safety regulations. Also, modifications of runway assignments and aircraft sequences are usually impossible if arriving aircraft have entered their final landing trajectory or if departing aircraft are lined up in the aircraft queue waiting for departure. Note that, depending on the airport, this freeze period in which runway sequences remain fixed can be shorter or longer. For departing aircraft, a target take-off time can usually be determined once the aircraft arrived at the departure gate. Given typical turn-around times, this is 90 to 120 minutes ahead of departure for wide-body aircraft serving long-haul destinations and 45 to 60 minutes ahead of departure for narrow-body aircraft serving domestic routes. Target take-off times become more accurate and reliable over time. Our proposed method, which recomputes optimal schedules every minute, benefits from this increasing accuracy. For an arriving aircraft, a reliable target landing time can be predicted once the aircraft is airborne. In both cases, we consider a deviation from the target time as delay, which induces aircraft-specific cost. Our goal is to minimize the sum of these delay cost over all aircraft and, thus, to minimize the overall cost of the runway schedule.

Each aircraft take-off and landing has to be scheduled within a certain time window. In general, an arriving aircraft can land within a time window where the earliest possible landing time is defined by the aircraft's target landing time, while available fuel limits the aircraft's latest possible landing time. Similar principles apply for departing aircraft, for which the earliest possible completion time of ground operations defines the push-back target time and earliest departure time. The latest possible departure time is theoretically unrestricted with later departures causing ever increasing delay cost.

Aircraft scheduled on the same runway have to follow certain separation requirements to comply with safety regulations imposed by aviation authorities, such as the Federal Aviation Administration (FAA) or the International Civil Aviation Organization (ICAO). Required separation times mainly depend on the wake vortices caused by the leading aircraft and their impact on following trailing aircraft. We consider separation requirements for all pairs of aircraft on the same runway, which Beasley et al. [4] defined as complete separation. Influencing factors for these separation requirements between aircraft are

- their relative positioning (leading or trailing),
- their operation modes (take-off or landing), and
- their weight class (large (e.g., Boeing 737, Airbus A320), heavy (e.g., Boeing 777, Airbus A340), or super (e.g., Airbus A380)).

Table 1 shows separation requirements for all combinations of aircraft operation classes usually operated at large international airports. Note that Boeing 757 aircraft are handled as "heavy" if leading and as "large" if trailing and, thus, build an own operation class with regard to separation requirements [5]. We implicitly assume that separation requirements are only dependent on operation classes. In practice, however, additional aspects such as standard instrument departure routes or enforced speed separations can influence required separation times between aircraft. While the model presented in this paper allows different separation times between all pairs of aircraft, a large number of different separation times increases the number of operation classes and, therefore, limits the effectiveness of pruning during preprocessing. The data of Table 1 also show that alternating landing and departing aircraft on the same runway (mixed runway operations) often leads to smaller separation times and, therefore, to runway schedules with higher throughput. Airports with only a few runways or operating at their capacity limit often use a mixed runway operation mode to increase the capacity of their runway systems. In general, such mixed runway operations increase the complexity of the scheduling problem and, therefore, the need for a mathematical optimization approach.

Models assigning runways and take-off or landing times to aircraft under these requirements in order to minimize delay have been developed and described in the academic literature. All of these models implicitly assume that runways are constantly avail-

Table 1
Separation requirements according to [5] (in seconds).

Leading	Trailing							
	Landing				Take-off			
	Large	Boeing 757	Heavy	Super	Large	Boeing 757	Heavy	Super
Landing								
Large	69	69	60	60	75	75	75	75
Boeing 757	157	157	96	96	75	75	75	75
Heavy	157	157	96	96	75	75	75	75
Super	180	180	120	120	180	180	120	120
Take-off								
Large	60	60	60	60	60	60	60	60
Boeing 757	60	60	60	60	120	120	90	90
Heavy	60	60	60	60	120	120	90	90
Super	180	180	120	120	180	180	120	120

able during the planning horizon or that periods of runway (un-) availability are known as an input parameter to the model in advance, e.g., if changing runway configurations are considered in the model formulation [6].

These assumptions do not hold during snowfall when runways have to be intermittently closed for snow removal or runway de-icing. In winter operations, snow removal on runways allows for certain flexibility regarding its precise timing. Hence, the start times of snow removals for different runways are subject to planning and scheduling itself. If snow starts piling up on a runway, flight operations continue as long as safe flight operations can be ensured. As soon as safe flight operations cannot be guaranteed anymore, the runway has to be closed temporarily and can only be reopened after the completion of a snow removal. When a runway is closed, aircraft have to be delayed or reassigned to other runways. We assume that, due to advanced weather predictions and a close monitoring of the runway conditions, the exact point in time at which runways become unsafe can accurately be forecast up to two hours and is known at the beginning of the operational planning horizon. After beginning snowfall, flight operations usually become unsafe at all runways of an airport at approximately the same time. Under continuous winter operations, i.e., if it is snowing for a longer period of several hours, the time at which flight operations on a runway become unsafe usually depends not only on the current snowfall but also on the elapsed time since the last snow removal on that runway.

For snow removal and runway de-icing, airports use dedicated snow removal groups. These snow removal groups consist of multiple snowplows and trucks, which clear a runway collaboratively in a coordinated performance. At most larger airports, snow removal equipment and their operators are a scarce resource and the number of runways exceeds the number of snow removal groups. Thus, not all runways can be cleared at the same time and snow removal groups have to clear multiple runways sequentially. The snow removal planning for multiple runways becomes a scheduling problem itself. Snow removal schedules have to consider snow removal durations per runway and transfer times between runways. In practice, clearing a runway usually takes around 20 minutes. The exact snow removal duration depends on the runway length and the used snow removal equipment. Additionally, snow removal groups need a certain amount of (setup) time between snow removals to drive from one runway to the next runway. These transfer times between runways depend on the physical layout of the airport and its road network.

After a runway has been cleared by a snow removal group, it usually stays safe for at least two hours. Hence, each runway has to be cleared not more than once within the considered planning horizon of up to two hours. Snow removal operations cause related cost, e.g., labor cost, depreciation of equipment, and cost for consumables. However, these cost are independent from the employed snow removal schedule and from sequencing decisions on aircraft and snow removals. From the perspective of the airport operator, snow removal cost are mainly fixed cost and, thus, can be neglected for making operational scheduling decisions.

We limit our work to the consideration of runway-related winter operations, i.e., snow removal activities on all runways of the airport. During snowfall, also taxiways, the apron, and all roads and driveways have to be cleared from snow and ice. Snow removal on this infrastructure, however, is often a subordinated planning task and conducted by separate snow removal groups using different equipment, e.g., smaller snow plows and other de-icing liquids or gravel. Also, aircraft de-icing is not explicitly modeled in our problem formulation but considered to be part of ground operations and reflected in take-off time windows for aircraft.

The integration of snow removal planning and aircraft scheduling constitutes a new problem class since scheduling snow removals differs structurally from scheduling aircraft with regard to resource demand, separation times, and the impact on runway availability. While aircraft require safe runways, snow removals require available snow removal groups and runways which can either be safe or unsafe. Sequence-dependent separation times between two aircraft apply for aircraft on the same runway. Sequence-dependent setup times between two snow removals, however, apply for snow removals on different runways conducted by the same snow removal group. Finally, snow removals impact whether or not runways can safely be operated.

3. Related literature

In this section, we briefly review related literature and research articles. First, we present general papers and model formulations for the *runway scheduling problem (RSP)*. Then, we focus on exact and heuristic solution approaches for the RSP. We briefly discuss related problems in the field of machine scheduling. We also review papers which consider other aspects of winter operations at airports but do not solve the runway scheduling problem itself. Finally, we give an overview of established pruning rules and valid inequalities for the RSP. For an extensive overview of related literature, we refer to [7,8]. Bennel et al. [7] presented a holistic literature review on optimizing runway schedules considering articles published until 2009. A concise and comprehensive overview of recent articles until 2015 can be found in [8].

General papers and model formulations The RSP, sometimes referred to as aircraft scheduling problem (ASP) or aircraft landing problem (ALP), has been widely studied. First models for the single runway case were proposed by Dear [9] and Psaraftis [10]. From there on, different model formulations and solution methodologies were developed ranging from very general frameworks [6] to specific applications [11,12]. The model by Beasley et al. [4] is one of the most cited and the basis for many subsequent problem formulations. The authors presented a mixed-integer model and a branch-and-bound solution approach for single and multiple runways. Herein, they minimize the sum of weighted earliness and delay, considering the possibility to schedule aircraft before their target time and assuming complete separation. For a stochastic version of the RSP, Solak et al. [13] proposed two different model formulations and a solution methodology based on sample average approximation and Lagrangian-based scenario decomposition.

Exact approaches To solve the RSP to optimality, various exact solution methods have been explored. Ernst et al. [14] proposed a branch-and-bound algorithm with a specialized simplex method. Samà et al. [15] formulated the ASP via an alternative graph formulation and presented a rolling horizon approach based on (FCFS) heuristics and exact (branch-and-bound) algorithms. D'Ariano et al. [16] solved the ASP minimizing the maximum consecutive delay over all aircraft with a branch-and-bound algorithm. MIP formulations for multiple different objective functions considering consecutive delay, total travel time spent, the number of deadline violations, aircraft priorities and equities, as well as completion times have been presented in [17,18]. In [19,20], the authors integrated the ASP with taxiway decisions under various objective functions and presented corresponding exact and heuristic solution approaches. Ghoniem et al. [21] presented advanced model formulations, e.g., an asymmetric traveling salesman problem with time windows, as well as efficient preprocessing procedures. Avella et al. [22], Faye [23], and Heidt et al. [24] presented time-indexed formulations. Ghoniem et al. [25] suggested a branch-and-price algorithm for the multiple runway problem. Dynamic programming approaches have been developed by Bennell et al. [26] for the single runway problem and by Lieder & Stolletz [8] for multiple interdependent and heterogeneous runways. Balakrishnan & Chandran [11] presented dynamic programming algorithms to solve the single runway problem under consideration of constrained position shifting prohibiting large deviations from a FCFS order.

Heuristic approaches Bianco et al. [27] showed that the general case of the RSP is NP-hard. Hence, many (meta-)heuristics have been developed to solve the RSP and related problems. These include local search algorithms [28,29], variable neighborhood, adaptive large neighborhood, and tabu search methods [12,30–33], simulated annealing [26,34], ant colony optimization [35], and population based heuristics [36–38].

Related machine scheduling problems The RSP can be seen as a machine scheduling problem with sequence-dependent processing or setup times where runways correspond to machines, aircraft correspond to jobs, and separation requirements correspond to sequence-dependent setup times. For the parallel machine scheduling problem with sequence-dependent setup times, Lee & Pinedo [39] proposed a heuristic based on dispatching rules and simulated annealing to minimize weighted tardiness. For the same problem and the objective of minimizing weighted earliness and tardiness, Radhakrishnan & Ventura [40] presented a simulated annealing approach. In order to minimize makespan, Vallada & Ruiz [41] suggested a genetic algorithm and Lopes & de Carvalho [42] developed a branch-and-price algorithm to minimize an additive cost function. An overview of scheduling problems with sequence-dependent setup times and related articles until 2008 was compiled by Allahverdi et al. [43]. From a machine scheduling point of view, snow removal has structural similarities to preventive maintenance, e.g., when maintenance crews have to maintain machines within a given time window and when there are travel times between machines for these crews moving from one machine to the next machine. Scheduling problems with maintenance activities have gained some interest in recent years. For the general problem of scheduling jobs and maintenance activities on parallel machines and the objective of minimizing total weighted completion time, Lee & Chen [44] presented a branch-and-bound algorithm based on column generation. Variants of this problem with aging effects, multiple maintenance activities during the planning horizon, or maintenance activities with start-time dependent durations have been investigated by Cheng et al. [45] and Yang et al. [46]. Ma et al. [47] presented a survey of scheduling problems with deterministic machine availability constraints considering articles until 2009. Gao et al. [48] studied the flexible job shop scheduling problem considering maintenance activities with flexible starting times and proposed a hybrid genetic algorithm to solve it. For a similar problem with scarce maintenance resources where only one machine can be maintained at any given time, Wang & Yu [49] proposed a heuristic algorithm based on a filtered beam search. Yoo & Lee [50] investigated different objective functions and job characteristics for the parallel machine scheduling problem with maintenance activities and developed dynamic programming algorithms to solve restricted cases of the problem. Paper integrating sequence-dependent setup times and machine maintenance usually consider makespan related objectives and solve the problem heuristically. For the single machine scheduling problem with sequence-dependent setup times and machine maintenance, Ángel-Bello et al. [51] proposed a heuristic based on a greedy construction algorithm and a subsequent local search improvement procedure. For a similar problem with dynamic job arrivals, Kaplanoğlu [52] presented a multi-agent based algorithm. Gholami et al. [53] developed a genetic algorithm for a flow shop problem with sequence-dependent setup times and random machine breakdowns. Heuristic methods to integrate jobs with sequence-dependent setup times and maintenance activities on multiple machines, specifically a genetic algorithm, a simulated annealing approach, and a variable neighborhood search, were presented in [54,55].

Winter operations at airports Regarding winter operations at airports, only a few publications exist. Janic [56] explicitly investigated the impact of heavy snowfall on airport operations using two deterministic queuing models. The first queuing model represents the accumulation of snow and ice on the airport surface and its impact on the airport's capacity while the second queuing model represents the handling of flights under such decreased airport capacity. Kotas & Bracken [57] presented a mixed-integer linear program to reschedule an airline's flight plan for a given day when the de-icing of aircraft causes delays and potential cancellations.

Pruning rules and valid inequalities for the RSP Maere et al. [58] gave a comprehensive overview of pruning rules for the RSP. The authors presented various generic pruning principles and embedded these pruning rules into a dynamic programming approach for the RSP. Valid inequalities for the RSP have been used by different authors to strengthen the linear relaxations of the model formulations. Beasley et al. [4] presented a basic set of valid inequalities for the single and multiple runway scenario. Ghoniem et al. [21] derived valid inequalities to tighten their problem formulation, which models the single runway case as an asymmetric traveling salesman problem with time windows.

Concluding, existing literature focuses on a wide range of aspects of runway scheduling. Most similar to our work are the paper of Lieder & Stolletz [8], in which the authors solve the deterministic RSP for landing and starting aircraft considering interdependent and heterogeneous runways via an exact dynamic programming approach, and the recent research of Samà et al. [18–20] focusing on the integration of aircraft trajectory optimization with

Table 2
Mathematical notation.

Sets and parameters	
$a \in \mathcal{A}$	Set of aircraft
T_a	Target take-off or landing time of aircraft a
L_a	Latest possible take-off or landing time of aircraft a
C_a	Cost coefficient per time unit for scheduling aircraft a after T_a
S_{ab}	Sequence-dependent separation time between aircraft a and b if a is scheduled before b on the same runway
O_a	Required separation time between aircraft a and a following snow removal on the same runway
$g \in \mathcal{G}$	Set of snow removal groups
$r \in \mathcal{R}$	Set of runways
U_r	Time at which flight operations on runway r become unsafe
P_r	Required time for a snow removal group to clear runway r
Q_{rs}	Sequence-dependent setup time between starts of snow removals on runways r and s conducted by the same snow removal group (including snow removal time P_r and required transfer time from runway r to runway s)
$i \in \mathcal{A} \cup \mathcal{R}$	Set of activities (aircraft and snow removals)
Decision variables	
$x_a \geq 0$	Take-off or landing time of aircraft a
$v_r \geq 0$	Start time of snow removal on runway r
δ_{ij}	$= \begin{cases} 1 & \text{if activity } i \text{ starts before activity } j \\ \in \{0,1\} & \text{otherwise} \end{cases}$
y_{ar}	$= \begin{cases} 1 & \text{if aircraft } a \text{ is scheduled on runway } r \\ 0 & \text{otherwise} \end{cases}$
z_{ab}	$= \begin{cases} 1 & \text{if aircraft } a \text{ and } b \text{ are scheduled on the same runway} \\ 0 & \text{otherwise} \end{cases}$
ρ_{rg}	$= \begin{cases} 1 & \text{if snow removal on runway } r \text{ is conducted by snow removal group } g \\ 0 & \text{otherwise} \end{cases}$
ϕ_{rs}	$= \begin{cases} 1 & \text{if snow removals on runways } r \text{ and } s \text{ are conducted by the same snow removal group} \\ 0 & \text{otherwise} \end{cases}$

aircraft sequencing under multiple objectives. However, the RSP during winter operations has not been investigated so far. In the following, we contribute to the research by presenting an efficient solution methodology for this problem using an exact optimization approach and applying pruning rules during preprocessing, valid inequalities for tightening the LP relaxation, and a heuristic to obtain an initial start solution. Our mathematical model builds upon the model in [4], which already supports optimal time-continuous solutions, complete separation, and multiple runways. We extend this model formulation to consider also snow removals and potential unavailabilities of runways.

4. Solution methodology

In this section, we detail and explain our solution methodology. In Section 4.1, we introduce a mathematical model for the RSP during winter operations. In Section 4.2, we describe three types of pruning rules; one established type and two novel types. To derive better LP bounds, we present valid inequalities in Section 4.3. Additionally, in Section 4.4, we develop a start heuristic for the MIP solver.

4.1. Mathematical model formulation

Using the notation introduced in Table 2, our model reflects the problem setting described in Section 2. In particular, our formulation assigns heterogeneous runways $r \in \mathcal{R}$ and take-off or landing times x_a to departing and arriving aircraft $a \in \mathcal{A}$. Further, it assigns homogeneous snow removal groups $g \in \mathcal{G}$ and snow removal times v_r to runways $r \in \mathcal{R}$. We assume mixed runway operations, where take-offs and landings can be scheduled on the same runway. The objective of our model is to minimize the total aircraft delay cost $\sum_a C_a(x_a - T_a)$ as the weighted deviations of the scheduled times x_a from the target times T_a.

We consider time windows $T_a \leq x_a \leq L_a$ for aircraft a limited by target times T_a and latest possible take-off or landing times L_a. Thus, aircraft are not allowed to be scheduled before their target time and we implicitly assume that earliest possible take-off or landing times are set as target times. We consider sequence-dependent separation requirements S_{ab} between pairs of aircraft a and b and separation requirements O_a between aircraft a and following snow removal activities. O_a is mainly defined by the duration of the take-off or landing procedure. Additionally, we consider runway closings in case too much snow, ice, or slush has piled up on a runway and safe operations cannot be guaranteed anymore. Once a runway has become unsafe at time U_r, it must be cleared before it can be reopened. If a runway r is cleared before time U_r, it stays safe for the remaining planning horizon. Our model formulation schedules exactly one snow removal of duration P_r on each runway r. If a runway does not become unsafe or if a snow removal on a runway is not necessary within the planning horizon due to low aircraft traffic, our formulation schedules a dummy snow removal at the end of the planning horizon. For snow removal groups, we consider sequence-dependent setup times Q_{rs} between runways r and s, which, for the ease of notation, include the time P_r to clear runway r and the transfer time from runway r to runway s. In our model, we use the concept of activities, which can be aircraft or snow removals, and their relative orders δ_{ij} with $i, j \in \mathcal{A} \cup \mathcal{R}, i \neq j$ to indicate whether i or j starts earlier.

Our MIP formulation is as follows:

$$\textbf{minimize} \quad \sum_{a \in \mathcal{A}} C_a(x_a - T_a) \tag{1}$$

subject to

$$T_a \le x_a \le L_a \qquad \forall\, a \in \mathcal{A} \tag{2}$$

$$\delta_{ij} + \delta_{ji} = 1 \qquad \forall\, i, j \in \mathcal{A} \cup \mathcal{R} : i \neq j \tag{3}$$

$$\sum_{r \in \mathcal{R}} y_{ar} = 1 \qquad \forall\, a \in \mathcal{A} \tag{4}$$

$$z_{ab} \ge y_{ar} + y_{br} - 1 \qquad \forall\, a, b \in \mathcal{A} : a \neq b; r \in \mathcal{R} \tag{5}$$

$$x_b \ge x_a + S_{ab} z_{ab} - M\delta_{ba} \qquad \forall\, a, b \in \mathcal{A} : a \neq b \tag{6}$$

$$\sum_{g \in \mathcal{G}} \rho_{rg} = 1 \qquad \forall\, r \in \mathcal{R} \tag{7}$$

$$\phi_{rs} \ge \rho_{rg} + \rho_{sg} - 1 \qquad \forall\, r, s \in \mathcal{R} : r \neq s; g \in \mathcal{G} \tag{8}$$

$$v_s \ge v_r + Q_{rs}\phi_{rs} - M\delta_{sr} \qquad \forall\, r, s \in \mathcal{R} : r \neq s \tag{9}$$

$$v_r \ge x_a + O_a y_{ar} - M\delta_{ra} \qquad \forall\, r \in \mathcal{R}; a \in \mathcal{A} \tag{10}$$

$$x_a \ge v_r + P_r y_{ar} - M\delta_{ar} \qquad \forall\, r \in \mathcal{R}; a \in \mathcal{A} \tag{11}$$

$$x_a \le U_r + M(1 - y_{ar}) + M\delta_{ra} \qquad \forall\, r \in \mathcal{R}; a \in \mathcal{A} \tag{12}$$

$$x_a, v_r \ge 0 \qquad \forall\, a \in A; r \in \mathcal{R}$$

$$\delta_{ij}, y_{ar}, z_{ab}, \rho_{rg}, \phi_{rs} \in \{0, 1\} \qquad \forall\, i, j \in \mathcal{A} \cup \mathcal{R} : i \neq j; a, b \in \mathcal{A} : a \neq b; r, s \in \mathcal{R} : r \neq s; g \in \mathcal{G}$$

The Objective (1) minimizes the sum over all aircraft's weighted delay. Constraints (2) secure that all flights are scheduled within their allowed time windows. Constraints (3) sequence all activities by deciding for each pair of activities i and j if i precedes j or vice versa. Our model formulation and definition of decision variables δ_{ij} also allow that activities can start at the same time if they are scheduled on different runways. Constraints (4) secure that each aircraft is scheduled on exactly one runway. Constraints (5) determine whether two aircraft are scheduled on the same runway. Constraints (6) guarantee that all separation requirements between aircraft on the same runway are met, imposing complete separation. Constraints (7) assign exactly one snow removal group to each runway. Constraints (8) determine whether two runways are cleared by the same snow removal group. Constraints (9) secure sufficient setup time between two consecutive snow removals conducted by the same group. Constraints (10) secure sufficient separation time between an aircraft and a following snow removal on the same runway. Similarly, Constraints (11) secure sufficient separation time between a snow removal and a following aircraft on the same runway. Finally, Constraints (12) make sure that an aircraft is only scheduled on a runway as long as flight operations are safe, i.e., before a runway becomes unsafe or after snow removal on that runway has been completed.

Note that runways can be heterogeneous in the sense that specific combinations of aircraft operations and runways can be prohibited. This allows to model cases in which specific runways are too short for aircraft of classes “heavy” or “super” or situations in which only take-offs or landings are permitted on certain runways. Fixing variables y_{ar} prohibits or enforces such assignments of aircraft a to runways r and decreases the complexity of the model since it prunes the branch-and-bound search tree. We assume that snow removal groups are identical in terms of driving speed and snow removal speed and, therefore, interchangeable. Nevertheless, it is possible to prohibit or enforce the assignments of snow removal groups g to runways r by fixing variables ρ_{rg} accordingly.

The objective function assumes linearly increasing cost for aircraft delay. Without significant modifications to the model, however, it is possible to extend the model formulation to a piecewise linear convex objective function. Start-time-dependent cost for snow removal operations could be reflected in the model by adding corresponding cost terms depending on variables v_r to the objective function. To incorporate fairness aspects, the model formulation can be extended with additional constraints, e.g., limiting the maximum delay per aircraft.

In our model formulation, we use big-M constraints to represent logical dependencies between variables. Big-M formulations typically enable compact mathematical models but regularly fail to provide strong LP bounds. This is unfavorable for the branch-and-bound procedure since it reduces the options to prune sub-optimal branches of the solution tree. Large instances of big-M formulations often become computational intractable. Too minimize the negative effects of big-M constraints, we choose them as small as possible. In our model, all big-M coefficients are sufficiently large if $M = \max_{a \in \mathcal{A}}\{L_a + O_a\} + r(\max_{r,s \in \mathcal{R}}\{Q_{rs}\} + \max_{r \in \mathcal{R}}\{P_r\})$ holds.

4.2. Problem-specific pruning rules

We apply pruning rules during preprocessing. We precalculate and fix binary variables to zero or one in order to decrease the solution space and to accelerate the branch-and-bound process. Maere et al. [58] give a comprehensive overview of various pruning rules for the runway scheduling problem. Most of these pruning rules are suitable for dynamic programming approaches, while only a few are also applicable for mixed-integer programs.

In the following, we discuss three types of pruning rules. While pruning rules of type I are established in literature, pruning rules of type II and III are novel. Pruning rules of type I order aircraft according to their non-overlapping time windows and were presented in [4]. Pruning rules of type II order aircraft of the same aircraft class and with overlapping time windows according to their target time. They are an extension of a pruning rule developed in [59]. Pruning rules of type III precalculate a dominant, i.e., optimal, snow removal pattern independently from actual aircraft traffic for many instances with homogeneous runways.

Pruning Rules of Type I: Strict Orders of Aircraft Based on Time Windows

Considering time windows as hard constraints, for some pairs of aircraft a and b, an optimal order can be determined based on their time windows (see Theorem 1). This was first proven in [4].

Theorem 1. *If the time window $[T_a, L_a]$ of aircraft a and the time window $[T_b, L_b]$ of aircraft b do not overlap with $L_a < T_b$, then aircraft a must be scheduled before aircraft b:*

$$\delta_{ab} = 1 \;\forall\, a, b \in \mathcal{A} : L_a < T_b$$

Pruning Rules of Type II: Strict Orders Within Aircraft Classes of Separation Identical Aircraft

Pruning rules of type II are based on the notion of aircraft classes.

Definition 1. *Aircraft classes of separation identical aircraft.* Aircraft a and a' are separation identical if they have the same pairwise separation requirements in relation to all other aircraft $b \in \mathcal{A} \setminus \{a, a'\}$, i.e., $S_{ab} = S_{a'b} \wedge S_{ba} = S_{ba'}\ \forall\, b \in \mathcal{A} \setminus \{a, a'\}$. A maximum set of pairwise separation identical aircraft constitutes an aircraft class.

For the objective of minimizing makespan and weighted delay, Psaraftis [10] showed that, within an aircraft class, a complete order can be inferred under the assumptions that no time window restrictions exist and that all aircraft of the set have the same cost function. Briskorn & Stolletz [59] showed that such complete orders within aircraft classes also exist if a time window order (see Definition 2) exists for all pairs of aircraft within a class and if class-specific piecewise linear convex cost functions are assumed.

Definition 2. *Time window order.* Two aircraft a and a' of the same aircraft class have a time window order $a \prec a'$ if $T_a < T_{a'} \Rightarrow L_a \leq L_{a'} \wedge L_a < L_{a'} \Rightarrow T_a \leq T_{a'}$.

Note that time windows $[T_a, L_a]$ and $[T_{a'}, L_{a'}]$ of aircraft a and a' can have different sizes. Briskorn & Stolletz [59] showed that scheduling two aircraft a and a' of the same class according to their time window order is optimal for minimizing delay cost by proving that swapping a and a' cannot improve the objective value. Different to [59], we relax the assumption of class-specific piecewise linear convex cost functions and assume aircraft-specific linear cost functions instead, i.e., we allow that aircraft a and a' of the same class have different cost coefficients $C_a \neq C_{a'}$. In practice, these cost coefficients often depend on the number of passengers aboard an aircraft and the price segment of the carrier (premium or low cost). In order to determine aircraft orders within aircraft classes, we introduce the concept of cost compliance.

Definition 3. *Cost compliance.* Two aircraft a and a' with time window order $a \prec a'$ are cost compliant if aircraft a has the same or a higher cost coefficient than aircraft a', i.e., $a \prec a' \Rightarrow C_a \geq C_{a'}$.

A strict order within a pair of aircraft a and a' of the same class can be precalculated if a time window order exists and the aircraft are cost compliant.

Theorem 2. *It is always optimal to schedule aircraft a and a' of the same class in their corresponding time window order if they are cost compliant:*

$$\delta_{aa'} = 1\ \forall\, a, a' \in \mathcal{A}: \quad \begin{aligned} & S_{ab} = S_{a'b} \wedge S_{ba} = S_{ba'}\ \forall\, b \in \mathcal{A} \setminus \{a, a'\}, \\ & T_a \leq T_{a'} \wedge L_a \leq L_{a'}, \\ & C_a \geq C_{a'} \end{aligned}$$

We present a proof for Theorem 2, which is based on a swap argument, in Appendix A.

Pruning Rules of Type III: Dominant Snow Removal Patterns for Homogeneous Runways

For a set of homogeneous runways (see Definition 4), it is often possible to precalculate a snow removal pattern (see Definition 5) which dominates all other possible snow removal patterns.

Definition 4. *Homogeneous runways.* We call a set of runways $\bar{\mathcal{R}} \subseteq \mathcal{R}$ homogeneous if all runways require the same amount of time for a snow removal activity ($P_r = P_{r'}\ \forall\, r, r' \in \bar{\mathcal{R}}$) and if they allow for the same operation modes, i.e., an aircraft that can take-off or land on a runway $r \in \bar{\mathcal{R}}$ can also take-off or land at all other runways $r' \in \bar{\mathcal{R}} : r' \neq r$.

Definition 5. *Snow removal pattern.* We refer to a snow removal pattern $\mathfrak{P}$ as an $|\mathcal{R}|$-tuple of triples (r_i, g_i, e_i), formally $\mathfrak{P} := \big((r_1, g_1, e_1), (r_2, g_2, e_2), \ldots, (r_{|\mathcal{R}|}, g_{|\mathcal{R}|}, e_{|\mathcal{R}|})\big)$. Herein, triple (r_i, g_i, e_i) defines the i-th snow removal activity in non-decreasing order of snow removal start times with $r_i \in \mathcal{R}$ denoting the runway being cleared, $g_i \in \mathcal{G}$ with $g_i = g : \rho_{r_i g} = 1$ denoting the assigned snow removal group, and e_i denoting the earliest possible start time of the i-th snow removal activity. We calculate earliest possible start times e_i based on preceding snow removals and sequence-dependent setup times Q_{rs}. Due to the order of the triples, $e_i \leq e_{i+1}\ \forall\, i \in 1, 2, \ldots, |\mathcal{R}| - 1$ holds.

Example 1. *Snow removal pattern* $\mathfrak{P} := \big((R2, SRG1, 0), (R3, SRG2, 0), (R1, SRG1, 2100)\big)$ describes a snow removal pattern where snow removal group SRG1 first clears runway R2 and then clears runway R1. Snow removal on R2 can start at time 0. Due to snow removal duration on R2 (1,200 seconds) and driving time from R2 to R1 (900 seconds), the earliest start time for snow removal on R3 is at 2,100. Snow removal group SRG2 clears runway R3 and can start this snow removal at 0.

Each snow removal pattern $\mathfrak{P}$ contains each runway $r \in \mathcal{R}$ exactly once since each runway is cleared exactly once within the planning horizon. Thus, only a finite number of different snow removal patterns exists. $\mathcal{P}$ denotes the finite set of all possible snow removal patterns and its cardinality is bounded by $|\mathcal{P}| = |\mathcal{R}|! \cdot |\mathcal{G}|^{|\mathcal{R}|}$.

Similar to the concept of active schedules [60], we introduce pseudo-active snow removal patterns.

Definition 6. *Pseudo-active snow removal pattern.* A snow removal pattern $\mathfrak{P}'$ is called pseudo-active if and only if no other snow removal pattern $\mathfrak{P}$ exists in which the i-th snow removal $(i = 1, 2, \ldots, |\mathcal{R}|)$ can be started (and, given homogeneous runways, also finished) earlier. Formally, $\mathfrak{P}' := \big((r'_1, g'_1, e'_1), (r'_2, g'_2, e'_2), \ldots, (r'_{|\mathcal{R}|}, g'_{|\mathcal{R}|}, e'_{|\mathcal{R}|})\big)$ is pseudo-active if $e'_i = \min_{\mathfrak{P} \in \mathcal{P}}\{e_i\}\ \forall\, i \in 1, 2, \ldots, |\mathcal{R}|$ holds.

Example 2. *Pseudo-active snow removal pattern*

- Consider a set of three possible snow removal patterns $\mathfrak{P}_1 := ((R2, SRG1, 0), (R3, SRG2, 0), (R1, SRG1, 2100))$, $\mathfrak{P}_2 := ((R2, SRG1, 0), (R3, SRG2, 0), (R1, SRG2, 2700))$, and $\mathfrak{P}_3 := ((R1, SRG1, 0), (R3, SRG2, 0), (R2, SRG1, 1800))$. Only $\mathfrak{P}_3$ is pseudo-active since, with $\mathfrak{P}_1$ or $\mathfrak{P}_2$, the first, the second, and the third snow removal cannot be started earlier. (With $\mathfrak{P}_3$, the third snow removal can start at time 1,800; with $\mathfrak{P}_1$, it can start only at 2,100 and, with $\mathfrak{P}_2$, only at 2,700.)
- Consider a set of three snow removal patterns $\mathfrak{P}_1 := ((R2, SRG1, 0), (R3, SRG2, 0), (R1, SRG1, 2100))$, $\mathfrak{P}_2 := ((R2, SRG1, 0), (R3, SRG2, 0), (R1, SRG2, 2700))$, and $\mathfrak{P}_3 := ((R1, SRG1, 0), (R3, SRG2, 1200), (R2, SRG1, 1800))$. No pattern is pseudo-active. $\mathfrak{P}_1$ is not pseudo-active since, with $\mathfrak{P}_3$, the third snow removal can be started earlier. $\mathfrak{P}_2$ is not pseudo-active since, with $\mathfrak{P}_3$, the third snow removal can be started earlier. $\mathfrak{P}_3$ is not pseudo-active since, with $\mathfrak{P}_2$, the second snow removal can be started earlier.

Extending a snow removal pattern with actual start times for each snow removal activity yields a snow removal schedule.

Definition 7. *Snow removal schedule.* A snow removal schedule $\mathfrak{S}$ extends a snow removal pattern and is a $|\mathcal{R}|$-tuple of quadruples $(r_i, g_i, e_i, \sigma_i)$, formally $\mathfrak{S} := ((r_1, g_1, e_1, \sigma_1), (r_2, g_2, e_2, \sigma_2), \ldots, (r_{|\mathcal{R}|}, g_{|\mathcal{R}|}, e_{|\mathcal{R}|}, \sigma_{|\mathcal{R}|}))$ where σ_i denotes the start time of the i-th snow removal.

Note that a snow removal schedule is feasible if and only if $e_i \leq \sigma_i\ \forall\, i \in 1, 2, \ldots, |\mathcal{R}|$ holds.

Example 3. *Snow removal schedule* $\mathfrak{S} := ((R2, SRG1, 0, 150), (R3, SRG2, 0, 0), (R1, SRG1, 2100, 2400))$ describes a snow removal schedule where snow removal group SRG1 first clears runway R2 at time 150 and then clears runway R1 at 2,400. Snow removal group SRG2 clears runway R3 at 0. A schedule $\mathfrak{S} := ((R2, SRG1, 0, 150), (R3, SRG2, 0, 0), (R1, SRG1, 2100, 2100))$ would not be possible since SRG1 would not have enough time between snow removals on R2 and R1 to clear R2 (1,200 seconds) and to drive from R2 to R1 (900 seconds).

A snow removal pattern is dominant if it can be extended to an optimal snow removal schedule and, thus, enables an optimal aircraft schedule. An optimal snow removal schedule must always consider aircraft traffic in the planning horizon and requires complete information about occurring aircraft. A dominant snow removal pattern, however, can often be precalculated in advance, independently from the occurring aircraft traffic. Finding a dominant snow removal pattern allows for fixing binary variables $\delta_{rs} \,\forall\, r, s \in \mathcal{R} : r \neq s$ and variable vectors $\boldsymbol{\rho}$ and $\boldsymbol{\phi}$ during preprocessing.

Theorem 3. *For homogeneous runways, a snow removal pattern* $\mathfrak{P}^* := ((r_1^*, g_1^*, e_1^*), (r_2^*, g_2^*, e_2^*), \ldots, (r_{|\mathcal{R}|}^*, g_{|\mathcal{R}|}^*, e_{|\mathcal{R}|}^*))$ *is dominant, i.e., it allows at least one snow removal schedule which is optimal, if*

1. *snow removal pattern* $\mathfrak{P}^*$ *is pseudo-active and*
2. *all runways are cleared in non-decreasing order with regard to their parameter* U_r.

Example 4. *Dominant snow removal pattern*

- Consider a set of three possible snow removal patterns $\mathfrak{P}_1 := ((R2, SRG1, 0), (R3, SRG2, 0), (R1, SRG1, 2100))$, $\mathfrak{P}_2 := ((R2, SRG1, 0), (R3, SRG2, 0), (R1, SRG2, 2700))$, and $\mathfrak{P}_3 := ((R1, SRG1, 0), (R3, SRG2, 0), (R2, SRG1, 1800))$ and times $U_1 = 500$, $U_2 = 2200$, and $U_3 = 1600$ at which runways become unsafe. $\mathfrak{P}_3$ is pseudo-active. With $\mathfrak{P}_3$, runways are cleared in order R1 $\rightarrow$ R3 $\rightarrow$ R2. Runways become unsafe in the same order ($U_1 \leq U_3 \leq U_2$). Hence, $\mathfrak{P}_3$ is dominant.
- Consider a set of three possible snow removal patterns $\mathfrak{P}_1 := ((R2, SRG1, 0), (R3, SRG2, 0), (R1, SRG1, 2100))$, $\mathfrak{P}_2 := ((R2, SRG1, 0), (R3, SRG2, 0), (R1, SRG2, 2700))$, and $\mathfrak{P}_3 := ((R1, SRG1, 0), (R3, SRG2, 0), (R2, SRG1, 1800))$ and times $U_1 = 3000$, $U_2 = 2200$, and $U_3 = 2200$ at which runways become unsafe. Only $\mathfrak{P}_3$ is pseudo-active. With $\mathfrak{P}_3$, runways are cleared in order R1 $\rightarrow$ R3 $\rightarrow$ R2. Runways become unsafe in a different order ($U_1 \not\leq U_3 \leq U_2$). Hence, $\mathfrak{P}_3$ is not dominant. No dominant snow removal patterns exists in this case.

We present a proof for Theorem 3 in Appendix A. Since the number of runways and snow removal groups at airports is rather small, we can enumerate all possible snow removal patterns in $\mathcal{P}$ in order to find a dominant pattern.

4.3. Problem-specific valid inequalities

In the following, we derive two new sets of valid inequalities specifically regarding snow removal activities. Although these constraints are redundant in the MIP formulation due to a combination of Constraints (2) and (12), they tighten the LP relaxation of the model.

Valid Inequalities I: Orders Between Snow Removals and Aircraft on the Same Runway

If an aircraft a is scheduled on runway r and the aircraft's target time T_a is later than the time U_r at which operations on runway r become unsafe, snow removal on runway r must be completed before aircraft a is scheduled and, thus, $\delta_{ra} = 1$. Consequently, for all pairs of aircraft a and runways r with $U_r < T_a$, the following inequality holds:

$$\delta_{ra} \geq y_{ar} \;\forall\, a \in \mathcal{A}; r \in \mathcal{R} : U_r < T_a \tag{13}$$

Valid Inequalities II: Global Orders Between Snow Removals and Aircraft

If the target time T_a of an aircraft a is later than all times U_r at which the different runways become unsafe, at least one snow removal has to be completed before aircraft a can be scheduled. Consequently, for all aircraft a with $\max_{r\in\mathcal{R}}\{U_r\} < T_a$, the following inequality holds:

$$\sum_{r\in\mathcal{R}} \delta_{ra} \geq 1 \;\forall\, a \in \mathcal{A} : \max_{r\in\mathcal{R}}\{U_r\} < T_a \tag{14}$$

4.4. Start solution heuristic

When solving a problem instance $\mathfrak{I}$, our proposed start solution heuristic generates a partial solution for $\mathfrak{I}$ from a related instance $\mathfrak{I}'$ which is designed to make maximal use of the proposed pruning rules and, thus, is efficiently solvable. To derive $\mathfrak{I}'$ from $\mathfrak{I}$, we fix all separation requirements S'_{ab} in $\mathfrak{I}'$ to the minimum of all separation requirements S_{ab} in $\mathfrak{I}$, i.e., $S'_{ab} = \min_{a,b\in\mathcal{A}:a\neq b}\{S_{ab}\} \;\forall\, a, b \in \mathcal{A} : a \neq b$, and we fix all cost factors to a constant c, i.e., $C_a = c \;\forall\, a \in \mathcal{A}$. Hence, in $\mathfrak{I}'$, all aircraft are separation- and cost-identical and pruning rules of type I and II (cf. Section 4.2) yield a complete order of all aircraft. This allows for solving $\mathfrak{I}'$ efficiently even for a large number of aircraft by applying the developed MIP (1)–(14). From the optimal solution of $\mathfrak{I}'$, we save the order of activities defined by variable vectors $\boldsymbol{\delta}'^*$ and the optimal snow removal schedule defined by variable vectors $\boldsymbol{\nu}'^*$ and $\boldsymbol{\rho}'^*$. To facilitate the construction of a first incumbent (start) solution for $\mathfrak{I}$, we forward this partial solution to the solver and initialize corresponding variable vectors $\boldsymbol{\delta}$, $\boldsymbol{\nu}$, and $\boldsymbol{\rho}$ in $\mathfrak{I}$ accordingly: $\boldsymbol{\delta} = \boldsymbol{\delta}'^*$, $\boldsymbol{\nu} = \boldsymbol{\nu}'^*$, and $\boldsymbol{\rho} = \boldsymbol{\rho}'^*$. In most cases, a good initial incumbent solution for $\mathfrak{I}$ can be constructed from this partial solution. In some cases, the partial solution cannot be completed to a valid start solution. In this case, the MIP solver cannot benefit from the partial solution and finds a first feasible incumbent solution during the branch-and-bound procedure. We provide the pseudo code of the proposed start solution heuristic in Appendix B.

5. Results

To show the applicability and effectiveness of our approach, we tested it on realistic data from Munich International Airport and on large-scale instances from the literature. In Section 5.1, we describe the setup and design of our computational study based on the realistic data set of Munich International Airport. We introduce a practice-based benchmark heuristic that mimicks the scheduling procedure applied manually by runway managers and air traffic controllers. We compare our exact approach with the practice-based heuristic on the realistic data set in Section 5.2 and demonstrate its applicability to larger problems in Section 5.3 by solving the large-scale aircraft landing instances introduced in [61]. Finally, we discuss some managerial remarks in Section 5.4.

5.1. Experimental design for the realistic data set of munich international airport

Data Set

We generated all instances using real arrival and departure data for a winter day at Munich International Airport. As a typical hub airport, Munich International Airport has time windows in which mainly domestic flights arrive and depart and time windows in which additional long-distance flights are handled. We considered three data sets with real flight operations recorded in November 2017, each with a representative mix of arriving and departing, domestic and long-distance flights. The data sets reflect different times of the day: 9 a.m. - 10:30 a.m. (data set "morning"), 12 p.m. - 1:30 p.m. (data set "noon"), and 9 p.m. - 10:30 p.m. (data set "evening").

Runways and Snow Removal Groups

We consider instances with either two runways and one snow removal group or with three runways and two snow removal groups. We assume sets of homogeneous and independent runways, i.e., that runways have the same length and allow for the same flight operations and that flight operations on one runway do not affect flight operations on other runways. Transfer times between runways are listed in Appendix C. For the first two runways, transfer times are based on the physical layout and road network of Munich International Airport. For instances with three runways, we consider an extension of the existing runway system as planned by the airport.

Aircraft and Flight Density

Instances of the realistic data set cover 30, 45, 60, or 75 aircraft to be scheduled. For instances with two runways and one snow removal group, we assume a flight density of 45 flight operations per hour. For instances with three runways and two snow removal groups, we consider 60 flight operations per hour.

Aircraft Time Windows and Delay Cost Factors

For all aircraft, we assume a maximum possible tardiness of 20 minutes. To take into account that delays of larger aircraft with a higher number of passengers are more critical, we use delay cost coefficients C_a of 1, 2, 3, and 4 monetary units for aircraft of the weight classes "large", "Boeing 757", "heavy", and "super" respectively.

Snowfall Scenarios

We distinguish between three scenarios which differ in their characteristics regarding weather and snowfall:

Scenario 1: Beginning snowfall At the start of the planning horizon, all runways have the same conditions, and, due to snowfall, operations become unsafe at the same point in time on all runways, i.e., $U_1 = U_2 = \ldots = U_{|\mathcal{R}|}$. We assume that runways become unsafe 25 minutes after the beginning of the planning horizon.

Scenario 2: Continuous winter operations Runways have previously been cleared from snow, ice, and slush at different times. Thus, the times at which runways become unsafe mainly depend on the times elapsed since the previous snow removals and, consequently, runways become unsafe at different times, i.e., $U_1 \neq U_2 \neq \ldots \neq U_{|\mathcal{R}|}$. We assume that the first runway becomes unsafe 10 minutes after the beginning of the planning horizon and that the second runway becomes unsafe after 25 minutes. In case of three runways, we assume that the third runway becomes unsafe 40 minutes after the beginning of the planning horizon.

Scenario 3: Ending snowfall Runways have previously been cleared and not all runways become unsafe during the planning horizon as snowfall is about to end. In case of three runways, two runways become unsafe (after 10 and 25 minutes). In case of two runways, one runway becomes unsafe (after 25 minutes). To reflect the lower demand for snow removals, we solved both cases using one snow removal group.

Considered Instances

For our computational study on the realistic data set, we consider 24 instances which reflect typical situations at a large airport during a winter day with snowfall. The instances combine different snowfall scenarios, times of day, runway configurations, and aircraft numbers in a realistic way. For Scenario 1 (beginning snowfall) we use the data set "morning", for Scenario 2 and Scenario 3, we use data sets "noon" and "evening" respectively. We compute all scenarios with two and three runways and for 30, 45, 60, and 75 aircraft. We provide a complete list of parameters per instance and supplementary instance data, such as required transfer times between runways, in Appendix C. We also provide all instance data online at Mendeley Data (see https://doi.org/10.17632/24nvbygjy7.2).

Naive Scheduling Approach

A simple and naive approach schedules aircraft and snow removals in parallel on all runways using a FCFS policy. Aircraft are scheduled in compliance with their time windows as early as possible and in ascending order of their target time. They are scheduled on the first runway which is or becomes available. Snow removals are scheduled on runways in the same order as these runways become unsafe. On each runway, the snow removal activity starts as soon as the runway is unsafe and a snow removal group is available. We consider this naive approach since it is the simplest and most obvious way to schedule aircraft and snow removals. Most airports, however, follow more advanced heuristic approaches yielding significantly better schedules.

Practice-based Benchmark Heuristic

To derive meaningful insights regarding the effectiveness of our solution approach, we compare our optimal schedules against schedules generated by a practice-based heuristic. This heuristic is based on discussions with decision makers of runway management and air traffic control at Munich International Airport. It simulates and mimicks the actual decision making process of human planners and serves as an approximation to decisions observed in practice. Therefore, we use schedules computed by the practice-based heuristic as benchmark solutions. Inspired by the human decision making process, the heuristic follows a stepwise and sequential approach:

Step 1 Schedule all snow removals within the planning horizon by sequentially applying the following two rules:

1. *Maximize runway availability* in order to increase the flexibility for scheduling aircraft. This is equivalent to minimizing the time spans in which runways are unavailable because they are unsafe or are being cleared.
2. *Start snow removal activities as late as possible* in order to delay the need for the next snow removals in the next planning period.

Step 2 Assign runways and take-off or landing times to aircraft by scheduling them in parallel on all runways on a FCFS basis. Again, aircraft are scheduled in compliance with their time windows as early as possible on the first available runway and in ascending order of their target time.

In our computational study, we additionally consider a variant of this heuristic in which we use the optimization model in [4] to schedule aircraft optimally in Step 2.

For more complex instances with two or more snow removal groups and three or more runways, the options to schedule snow removals rapidly increase. In these cases, the benchmark heuristic computes optimal solutions for the inherent maximization problem in Step 1, which are often difficult to find for human planners.

5.2. Computational results for the realistic data set

We computed all results on an Intel i7-8700K with 3.7 GHz and 32 GB RAM using Python 3.6 with Gurobi 8.1. When solving the

	Quartile				
	min	.25	**.50**	.75	max
Computational time (in seconds)	18.3	28.2	**39.7**	60.6	115.0

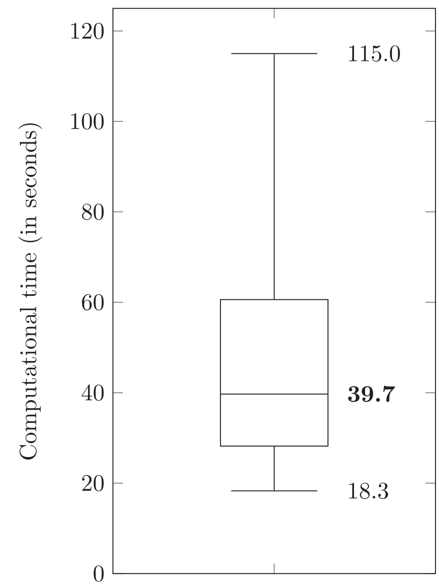

Fig. 1. Exemplary distribution of computational times for the same instance.

same problem instance multiple times, we observed a high variance in computational times. This variance results from the degeneracy in the LP relaxation of the MIP; if a problem's LP relaxation has multiple optimal solutions, the solver chooses one of these LP solutions randomly based on a random seed number. Resulting branching decisions change the search tree and lead to varying computational times. In order to derive meaningful conclusions about the performance of our approach, we solved each instance 60 times with different random seeds. Fig. 1 exemplary shows the distribution of computational times for an instance of Scenario 1 with 75 aircraft, two runways and one snow removal group using all pruning rules and valid inequalities. We observe that most computational times (between the 0.25-quartile and the 0.75-quartile) are closely centered around the median. For some random seeds, we notice considerably lower or higher computational times with upper outliers being more extreme.

In the following, we report the median of computational times. Table 3 shows the results for all computed instances of the realistic data set. For each instance, we report the objective function value z^{naive} of the naive scheduling approach and the objective function value $z^{\text{heu/FCFS}}$ of the practice-based heuristic as a benchmark solution. We also report the objective function value $z^{\text{heu/opt}}$ of a variant of the practice-based heuristic which, in Step 2, uses an optimization model to schedule aircraft optimally. With $\Delta^{\text{heu/opt}} = (z^{\text{heu/FCFS}} - z^{\text{heu/opt}})/z^{\text{heu/FCFS}}$, we measure the improvement through state-of-the-art aircraft scheduling over the FCFS-based aircraft scheduling approach used in practice given that snow removals are scheduled with the practice-based heuristic. We show the optimal objective function value z^{MIP} of our integrated solution approach and the improvement $\Delta^{\text{MIP}} = (z^{\text{heu/FCFS}} - z^{\text{MIP}})/z^{\text{heu/FCFS}}$ through our solution methodology over the practice-based heuristic. For our MIP formulation, we report computational times for four model configurations:

- Configuration C1: MIP as defined by (1)–(12)
- Configuration C2: MIP using pruning rules of type I and II
- Configuration C3: MIP using pruning rules of type I–III
- Configuration C4: MIP using pruning rules of type I–III and valid inequalities I and II

In our computational experiments, we found that good solutions are computed early during the branch-and-bound process. Therefore, in the last column, we report required times to compute solutions which deviate less than 1% from the optimal objective function value z^{MIP} using configuration C4 and our start solution heuristic. We report the computational time of the start heuristic in brackets.

Reduction of Delay Cost Through Integrated Planning

To demonstrate that an integrated planning of aircraft and snow removals significantly reduces delay cost, we compare delay cost of heuristic solutions z^{naive}, $z^{\text{heu/FCFS}}$, and $z^{\text{heu/opt}}$ with optimal delay cost z^{MIP}.

An analysis of z^{naive} shows that, for most instances, the naive scheduling approach constructs solutions significantly worse than all other considered approaches and is not suitable for an application in practice.

The comparison of the objective function values of the practice-based heuristic $z^{\text{heu/FCFS}}$ and the optimization model z^{MIP} shows that our integrated approach significantly reduces delay cost and yields improvements Δ^{MIP} of up to 58% (and 69% on average). These improvements reflect the benefit of an optimal aircraft schedule with integrated snow removal decisions. Improvements $\Delta^{\text{heu/opt}}$ show that using an optimal aircraft schedule instead of a FCFS-based aircraft schedule within the practice-based sequential heuristic reduces delay cost only up to 33% (and 18% on average). Comparing Δ^{MIP} and $\Delta^{\text{heu/opt}}$ indicates that, for many instances, substantial delay cost reductions originate from an integration of the scheduling decisions for snow removals and aircraft.

Improvements of Computational Times Through Pruning Rules, Valid Inequalities, and the Start Solution Heuristic

The proposed pruning rules, the valid inequalities, and the start solution heuristic significantly reduce computational times, which enables the application of the proposed model in real-world settings.

Our computational results show that for 13 out of 24 instances, the original MIP formulation (C1) does not yield a proven optimal solution within a time limit of one hour.

For all instances, we observe significant improvements in the computational time by using pruning rules. The application of

Table 3
Computational results for the realistic data set.

						Computational times of different configurations (in seconds)				
Instance	# aircraft / # runways / # snow removal groups	z^{naive}	$z^{\text{heu/FCFS}}$	$z^{\text{heu/opt}}$ / $\Delta^{\text{heu/opt}}$	z^{MIP} / Δ^{MIP}	C1 (MIP)	C2 (MIP + PR I and II)	C3 (MIP + PR I–III)	C4 (MIP + PR I–III + VI I and II)	Solution with less than 1% deviation from optimal value
Scenario 1: Beginning snowfall (data set "morning")										
1	30 / 2 / 1	15,757	2,415	1,726 / 29%	1,710 / 29%	41	1	0	0	0 (+0)
2	45 / 2 / 1	31,466	3,592	2,933 / 18%	2,933 / 18%	> 3,600	29	6	6	0 (+0)*
3	60 / 2 / 1	49,761	3,732	3,070 / 18%	3,070 / 18%	> 3,600	112	21	20	1 (+0)
4	75 / 2 / 1	62,083	3,804	3,142 / 17%	3,142 / 17%	> 3,600	206	44	40	2 (+0)
5	30 / 3 / 2	11,057	1,027	1,006 / 2%	1,006 / 2%	> 3,600	48	4	3	1 (+1)
6	45 / 3 / 2	25,492	3,655	2,825 / 23%	1,683 / 54%	> 3,600	> 3,600	627	601	2 (+2)
7	60 / 3 / 2	32,837	4,179	3,185 / 24%	1,747 / 58%	> 3,600	> 3,600	1,306	1,015	10 (+19)
8	75 / 3 / 2	38,476	4,198	3,201 / 24%	1,763 / 58%	> 3,600	> 3,600	1,728	1,680	16 (+28)
Scenario 2: Continuous winter operations (data set "noon")										
9	30 / 2 / 1	7,423	4,034	3,157 / 22%	3,157 / 22%	961	2	4	4	2 (+0)*
10	45 / 2 / 1	12,192	6,150	4,352 / 29%	4,352 / 29%	> 3,600	19	15	14	4 (+0)
11	60 / 2 / 1	15,690	6,451	4,522 / 30%	4,522 / 30%	> 3,600	42	43	45	7 (+0)
12	75 / 2 / 1	15,752	6,513	4,584 / 30%	4,584 / 30%	> 3,600	53	47	48	8 (+0)
13	30 / 3 / 2	1,225	499	499 / -	463 / 7%	34	2	2**	2**	0 (+0)**
14	45 / 3 / 2	2,653	1,268	1,266 / -	634 / 50%	1,994	29	29**	24**	1 (+0)**
15	60 / 3 / 2	2,901	1,529	1,491 / 2%	707 / 54%	> 3,600	95	95**	76**	3 (+3)**
16	75 / 3 / 2	2,982	1,610	1,548 / 4%	739 / 54%	> 3,600	216	216**	191**	14 (+4)**
Scenario 3: Ending snowfall (data set "evening")										
17	30 / 2 / 1	721	721	700 / 3%	700 / 3%	31	3	2	3	1 (+0)
18	45 / 2 / 1	1,400	1,400	1,176 / 16%	891 / 36%	203	7	7	7	1 (+0)
19	60 / 2 / 1	1,474	1,474	1,221 / 17%	936 / 36%	304	10	9	9	2 (+1)
20	75 / 2 / 1	1,819	1,819	1,566 / 14%	1,230 / 32%	> 3,600	309	221	210	11 (+2)
21	30 / 3 / 2	735	387	261 / 33%	261 / 33%	21	2	1	2	1 (+0)
22	45 / 3 / 2	838	490	364 / 26%	364 / 26%	219	25	6	7	2 (+1)
23	60 / 3 / 2	931	531	405 / 24%	405 / 24%	1,240	43	14	13	3 (+1)
24	75 / 3 / 2	960	560	418 / 25%	418 / 25%	1,879	53	20	19	4 (+1)

* heuristic does not yield a feasible start solutions (first feasible incumbent solution is created by MIP solver during the branch-and-bound procedure)
** no optimal snow removal pattern exists for this instance and, therefore, pruning rules of type III do not improve computational times
PR I, II, III = pruning rules of type I, II, III   VI I, II= valid inequalities I, II

Table 4
Individual contributions of pruning rules to improvements of computational times.

	Computational times (in percent of C1)
MIP without pruning rules (C1)	100%
MIP using pruning rules of type I	95%
MIP using pruning rules of type II	5%
MIP using pruning rules of type III	72% (66%*)

* for instances with dominant snow removal patterns

pruning rules of type I and II (C2) allows us to solve 21 of 24 instances to optimality within one hour.

Pruning rules of type III (C3), which compute an optimal snow removal pattern, allow us to solve all instances to optimality, 18 of them within one minute. Additionally, they reduce the computational times by up to 92% in Scenario 1 and by up to 76% in Scenario 3. In Scenario 2, optimal snow removal patterns only exist for instances with two runways and one snow removal group (Instances 9–12). In these cases, pruning rules of type III reduce computational times by up to 21%.

Table 4 shows the individual effects of all pruning rules on computational times. For each pruning rule, we report resulting average computational times as percentage of computational times for the MIP without pruning rules (C1). Pruning rules of type II yield the highest improvement, decreasing computational times, on average, to 5%. Also, pruning rules of type I and III have a positive effect and reduce computational times, on average, to 95% (pruning rules of type I) and 72% (pruning rules of type III). Note that Table 4 is based on all 11 instances for which the MIP (C1) can be solved within the time limit of one hour and, therefore, meaningful improvements of computational times can be calculated. The reported average values include two instances without optimal snow removal patterns (Instances 13 and 14), for which pruning rules of type III do not yield improvements. For instances with dominant snow removal patterns, pruning rules of type III reduce computational times to 66% on average.

For six instances (Instances 4, 5, 7, and 14–16), valid inequalities (C4) yield a significant speed-up of at least 10%. Especially for Instances 14–16, where pruning rules III do not yield an optimal snow removal pattern, the proposed valid inequalities improve computational times by up to 20%. For all six instances which we cannot solve to optimality within one minute, we compute schedules with less than 1% deviation from the optimal objective function value in less than a minute; this is also the case if the start heuristic does not yield a feasible solution. For 22 out of 24 instances, the start heuristic computes initial start solutions within a few seconds. In two cases (Instances 2 and 9), where the partial solution does not generate a feasible incumbent solution, the MIP solver cannot benefit from the proposed start heuristic and finds the first feasible incumbent solution during the branch-and-bound procedure. Note that this squanders the computational effort of the start heuristic but does not impact the general feasibility of the model. The start heuristic is particularly helpful to find good solutions early in the branch-and-bound procedure solving the MIP. As a result, it seems reasonable to use our presented approach also heuristically by terminating the branch-and-bound procedure before a proven optimum is obtained and using the incumbent at termination, i.e., the best solution found so far.

Concluding, the proposed model can be used in a real-world setting to compute optimal runway schedules for a planning

Table 5
Computational results for large-scale instances.

Instance	# aircraft / # runways / # snow removal groups	z^{MIP}	Computational times (in seconds) C4 (MIP + PR I–III + VI I and II)	Solution with less than 1% deviation from optimal value
airland9 instances				
25	100 / 3 / 2	109.43	1	1 (+0)
26	100 / 4 / 2	0	1	0 (+0)
airland10 instances				
27	150 / 3 / 2	263.15	> 3,600	3 (+0)
28	150 / 4 / 2	34.22	2	1 (+0)
29	150 / 5 / 3	0	1	0 (+0)
airland11 instances				
30	200 / 3 / 2	307.07	>3,600	3 (+1)
31	200 / 4 / 2	69.66	8	3 (+1)
32	200 / 5 / 3	0	2	2 (+1)
airland12 instances				
33	250 / 3 / 2	246.88	80	5 (+1)
34	250 / 4 / 2	2.86	3	2 (+1)
35	250 / 5 / 3	0	2	2 (+2)
airland13 instances				
36	500 / 3 / 2	778.60	>3,600	36 (+5)
37	500 / 4 / 2	90.76	122	18 (+5)
38	500 / 5 / 3	0	8	0 (+6)

PR I, II, III = pruning rules of type I, II, III   VI I, II= valid inequalities I, II

horizon of up to two hours. Since the model considers aircraft currently approaching the runways, a recalculation of the optimal schedule is necessary approximately once per minute, when the situation on the taxiways or in the airspace near the airport has changed, i.e., if aircraft have departed or landed or if new aircraft have entered the airspace near the airport. Our presented exact solution methodology computes optimal runway schedules in most cases within one minute. When computational times of the exact approach become prohibitively large, we suggest to use our approach heuristically to obtain very good runway schedules within a few seconds.

5.3. Computational results for large-Scale instances

To demonstrate that our approach is also suitable for large-scale problems, we applied our algorithm to the large-scale aircraft landing instances *airland9–13* introduced in [61]. These instances comprise up to 500 aircraft and are available online from OR-Library (see http://people.brunel.ac.uk/~mastjjb/jeb/orlib/airlandinfo.html). We used aircraft parameters, i.e., target landing time, latest landing time, cost factor for delay, and separation requirements, as given in the original data set. We computed all instances for up to five runways and up to three snow removal groups, assuming that every 20 minutes another runway becomes unsafe. We list all instance parameters as well as required transfer times between runways in Appendix C and provide these instances also in our Mendeley Data set.

In Table 5, we report results for the large-scale instances focusing on optimal objective function values z^{MIP} and computational times. We report computational times for configuration C4, using all discussed pruning rules and valid inequalities, and we present required times to compute solutions which deviate less than 1% from the optimal objective function value z^{MIP} (with computational times for the start heuristic in brackets). Our algorithm solves 11 out of 14 large-scale instances to optimality within the time limit of one hour. We solve nine instances to optimality within one minute. For all instances, we find solutions which deviate less than 1% from the optimal objective function value in less than one minute.

5.4. Managerial remarks

Our computational study indicates that an integrated planning of aircraft and snow removals on airport runways improves delay cost compared to simple scheduling rules which start snow removals as soon as runways are unsafe and snow removal groups are available.

Moreover, the results of our study can be used to synthesize the following two guiding principles for a manual or heuristic planning of snow removal schedules. First, we note that one should avoid schedules in which a runway has to be closed due to snow without having a snow removal group available. To avoid such situations, one should favor schedules in which a (preceding) snow removal on a runway already starts before operations on that runway become unsafe, such that a succeeding snow removal on the next runway can start without delay. Second, one may consider the concept of dominant snow removal patterns during manual or heuristic planning. Such optimal snow removal patterns can be calculated in advance and describe potential efficient orders of snow removals and assignments of snow removal groups to runways.

One practical comment on these findings is in order. Our comparisons and the related findings imply that we assume equal information between a manual planner and our optimization-based approach. In practice, a human planner may have additional (immediate) information that is not reflected in our optimization model. In this case, the schedules computed by our algorithm might not be optimal. Accordingly, one may see our discussion and utilize the presented approach not in competition to human planners but to assist and possibly improve their planning during standard every-day situations.

6. Conclusion

In this paper, we presented the first optimization model for the runway scheduling problem during winter operations. Our approach minimizes aircraft delay and associated cost through an integration of the scheduling decisions for aircraft and snow removals. Due to its computational efficiency, it can be utilized in practice to solve the runway scheduling problem with winter operations at large airports.

We modeled the problem as a mixed-integer linear program. In order to accelerate the branch-and-bound procedure, we derived problem-specific pruning rules and valid inequalities. Additionally, we presented a method to derive an initial start solution for the solver heuristically. To substantiate the value of our work, we applied the presented approach to realistic instances from an international airport and to large-scale instances from the literature. The numerical experiments showed significant reductions of delay cost compared to a manual scheduling process by a human planner. We showed that our approach can solve all 24 instances of the realistic data set optimally within a time limit of one hour and 18 instances optimally within one minute. For all instances, our approach yields very good solutions deviating less than 1% from the optimal objective function value in less than one minute. Hence, the presented method can also be used heuristically by terminating the computation early. Based on the computational study, we presented several managerial remarks.

The results of our work motivate some potential future research directions. This paper solved the static and deterministic version of the runway scheduling problem during winter operations. To address longer planning horizons, it is reasonable to investigate also the dynamic variant of the problem. Since many aspects of the problem setting, e.g., aircraft time windows, aircraft parameters, and weather conditions, are subject to uncertainty and change, it might be beneficial to account for the stochasticity of the problem. Related to that, concepts of robust optimization could help to create more stable schedules which are less likely to change with minor modifications of the parameters. Also, it might be possible to integrate the considered decision problems with adjacent tasks such as aircraft routing through the taxiways, aircraft de-icing, or snow removal on other parts of the airport. We believe that the proposed methodology can also be adapted to related problems in machine scheduling, especially if sequence-dependent setup times and maintenance activities are taken into account.

We are confident that the results of our work and of potential future research in the suggested directions enable airport operators and air traffic controllers to integrate their planning and, thus, to achieve better overall runway schedules for winter days with considerable snowfall.

Appendix A. Mathematical Proofs

In the following, we present proofs for Theorem 2 and Theorem 3.

Theorem 2. *It is always optimal to schedule aircraft a and a′ of the same class in their corresponding time window order if they are cost compliant:*

$$\delta_{aa'} = 1 \;\forall\, a, a' \in \mathcal{A}: \quad S_{ab} = S_{a'b} \wedge S_{ba} = S_{ba'} \;\forall\, b \in \mathcal{A}\setminus\{a, a'\}, \tag{15}$$

$$T_a \le T_{a'} \wedge L_a \le L_{a'}, \tag{16}$$

$$C_a \ge C_{a'} \tag{17}$$

Proof. We consider two cases:

Case 1: The time windows of aircraft a and a' do not overlap. In this case, scheduling a' before a is not feasible and a must be scheduled before a'. Note that Theorem 1 applies in this case.

Case 2: The time windows of aircraft a and a' overlap with $L_a \ge T_{a'}$. Consider a feasible aircraft schedule $\mathcal{S}$ with $x_a > x_{a'}$ (a' is scheduled before a) and assume time window order ($T_a \le T_{a'} \wedge L_a \le L_{a'}$) and cost compliance ($C_a \ge C_{a'}$). We proof that swapping a and a' cannot increase the objective value: We generate a new schedule $\bar{\mathcal{S}}$ where we swap a and a' and schedule aircraft a at $\bar{x}_a = x_{a'}$ and aircraft a' at $\bar{x}_{a'} = x_a$. This schedule $\bar{\mathcal{S}}$ is feasible with regard to other aircraft since a and a' are separation identical, and it is feasible with regard to possible time windows since $T_a \le T_{a'} \le x_{a'} = \bar{x}_a < x_a = \bar{x}_{a'} \le L_a \le L_{a'}$. From cost compliance (18) and $x_{a'} = \bar{x}_a < x_a = \bar{x}_{a'}$, we conclude by (19)–(24) that the objective value $O(\bar{\mathcal{S}}) = C_a(\bar{x}_a - T_a) + C_{a'}(\bar{x}_{a'} - T_{a'})$ of the new schedule $\bar{\mathcal{S}}$ where a and a' are swapped (and a is scheduled before a' according to their time window order) is better than or equal to the objective value $O(\mathcal{S}) = C_a(x_a - T_a) + C_{a'}(x_{a'} - T_{a'})$ of the original schedule $\mathcal{S}$:

$$C_a \ge C_{a'} \tag{18}$$

$$\Leftrightarrow \quad C_a(\bar{x}_a - x_a) \le C_{a'}(x_{a'} - \bar{x}_{a'}) \tag{19}$$

$$\Leftrightarrow \quad C_a\bar{x}_a - C_a x_a \le C_{a'}x_{a'} - C_{a'}\bar{x}_{a'} \tag{20}$$

$$\Leftrightarrow \quad C_a\bar{x}_a + C_{a'}\bar{x}_{a'} \le C_{a'}x_{a'} + C_a x_a \tag{21}$$

$$\Leftrightarrow \quad C_a\bar{x}_a + C_{a'}\bar{x}_{a'} - C_aT_a - C_{a'}T_{a'} \le C_{a'}x_{a'} + C_a x_a - C_aT_a - C_{a'}T_{a'} \tag{22}$$

$$\Leftrightarrow \quad C_a(\bar{x}_a - T_a) + C_{a'}(\bar{x}_{a'} - T_{a'}) \le C_a(x_a - T_a) + C_{a'}(x_{a'} - T_{a'}) \tag{23}$$

$$\Leftrightarrow \quad O(\bar{\mathcal{S}}) \le O(\mathcal{S}) \tag{24}$$

□

Theorem 3. *For homogeneous runways, a snow removal pattern $\mathfrak{P}^* := ((r_1^*, g_1^*, e_1^*), (r_2^*, g_2^*, e_2^*), \ldots, (r_{|\mathcal{R}|}^*, g_{|\mathcal{R}|}^*, e_{|\mathcal{R}|}^*))$ is dominant, i.e., it allows at least one snow removal schedule which is optimal, if*

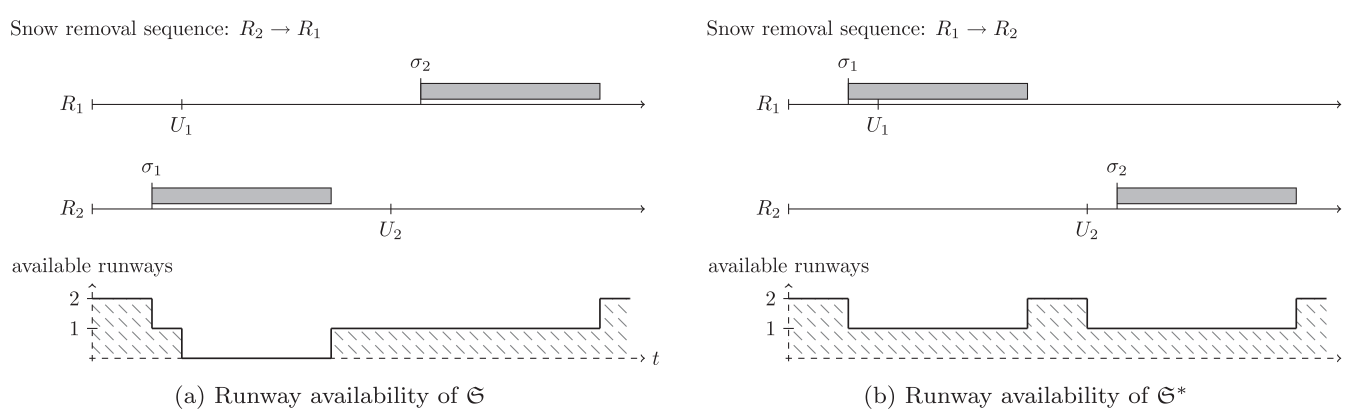

Fig. 2. Example: $\mathfrak{S}^*$ outperforms $\mathfrak{S}$ in terms of runway availability.

1. *snow removal pattern $\mathfrak{P}^*$ is pseudo-active and*
2. *all runways are cleared in non-decreasing order with regard to their parameter U_r.*

Proof. In the following, we show that we can always construct an optimal snow removal schedule $\mathfrak{S}^*$ by extending a pseudo-active snow removal pattern $\mathfrak{P}^* = ((r_1^*, g_1^*, e_1^*), (r_2^*, g_2^*, e_2^*), \dots, (r_{|\mathcal{R}|}^*, g_{|\mathcal{R}|}^*, e_{|\mathcal{R}|}^*))$ with $U_{r_1^*} \leq U_{r_2^*} \leq \dots \leq U_{r_{|\mathcal{R}|}^*}$ if runways are homogeneous. Consider a feasible and optimal snow removal schedule $\mathfrak{S} = ((r_1, g_1, e_1, \sigma_1), (r_2, g_2, e_2, \sigma_2), \dots, (r_{|\mathcal{R}|}, g_{|\mathcal{R}|}, e_{|\mathcal{R}|}, \sigma_{|\mathcal{R}|}))$. We construct a schedule $\mathfrak{S}^* = ((r_1^*, g_1^*, e_1^*, \sigma_1), (r_2^*, g_2^*, e_2^*, \sigma_2), \dots, (r_{|\mathcal{R}|}^*, g_{|\mathcal{R}|}^*, e_{|\mathcal{R}|}^*, \sigma_{|\mathcal{R}|}))$ extending dominant snow removal pattern $\mathfrak{P}^*$ with snow removal start times $\sigma_i \, \forall \, i = 1, 2, \dots, |\mathcal{R}|$ of the optimal snow removal schedule $\mathfrak{S}$. From schedule $\mathfrak{S}$ being feasible ($e_i \leq \sigma_i \, \forall \, i \in 1, 2, \dots, |\mathcal{R}|$) and pattern $\mathfrak{P}^*$ being pseudo-active ($e_i^* = \min_{\mathfrak{P} \in \mathcal{P}}\{e_i\} \, \forall \, i \in 1, 2, \dots, |\mathcal{R}|$), it follows that $e_i^* \leq e_i \leq \sigma_i \, \forall \, i = 1, 2, \dots, |\mathcal{R}|$ and, hence, that schedule $\mathfrak{S}^*$ is feasible. In schedule $\mathfrak{S}^*$, all homogeneous runways are cleared in non-decreasing order with regard to their parameter U_r. Thus, schedule $\mathfrak{S}^*$ has the same or a better runway availability profile than schedule $\mathfrak{S}$, i.e., at every point in time, schedule $\mathfrak{S}^*$ provides at least as many available runways than schedule $\mathfrak{S}$ (cf. Fig. 2). Consequently, $\mathfrak{S}^*$ allows the same aircraft schedule and, therefore, the same objective value than $\mathfrak{S}$ and is also optimal. □

Appendix B. Pseudo Code for the Start Solution Heuristic

The pseudo code of our start solution heuristic is shown as Algorithm 1.

Algorithm 1 Start Solution Heuristic.

Require: Instance $\mathfrak{I}$
create $\mathfrak{I}'$ as copy of $\mathfrak{I}$
for all $a, b \in \mathcal{A} : a \neq b$ **do**
  $S'_{ab} := \min_{a,b \in \mathcal{A}: a \neq b}\{S_{ab}\}$
solve $\mathfrak{I}'$ to derive optimal values $\boldsymbol{\delta'^*}$, $\boldsymbol{\nu'^*}$, $\boldsymbol{\rho'^*}$
initialize corresponding variables in $\mathfrak{I}$: $\boldsymbol{\delta} = \boldsymbol{\delta'^*}, \boldsymbol{\nu} = \boldsymbol{\nu'^*}, \boldsymbol{\rho} = \boldsymbol{\rho'^*}$
solve $\mathfrak{I}$

Appendix C. Supplementary Instance Data

Table 6 presents details for the instances of the realistic data set and Table 7 presents details for the large-scale instances.

Table 8 shows transfer times between pairs of runways. Transfer times between the first three runways are based on the physical layout of Munich International Airport and a planned extension of the existing runways system. For the remaining pairs of runways, we use similar transfer times between 10 and 25 minutes.

Table 6
Characteristics and parameters for instances of the realistic data set.

Instance	Scenario	Data set	# aircraft / # runways / # snow removal groups	# departing aircraft of class large / Boeing 757 / heavy / super | # landing aircraft of class large / Boeing 757 / heavy / super	Planning horizon $\max_{a \in \mathcal{A}}\{L_a\}$ (in seconds)	Unsafe times U_1 / U_2 / U_3 (in seconds)	Required snow removal times P_1 / P_2 / P_3 (in seconds)
1	*Beginning snowfall*	*"morning" / 9 a.m.*	30 / 2 / 1	14 / 0 / 2 / 0 | 12 / 1 / 1 / 0	3,664	1,500 / 1,500	1,200 / 1,200
2	*Beginning snowfall*	*"morning" / 9 a.m.*	45 / 2 / 1	19 / 0 / 2 / 0 | 21 / 1 / 2 / 0	4,789	1,500 / 1,500	1,200 / 1,200
3	*Beginning snowfall*	*"morning" / 9 a.m.*	60 / 2 / 1	25 / 1 / 2 / 0 | 28 / 1 / 3 / 0	5,593	1,500 / 1,500	1,200 / 1,200
4	*Beginning snowfall*	*"morning" / 9 a.m.*	75 / 2 / 1	31 / 2 / 2 / 0 | 35 / 1 / 4 / 0	6,608	1,500 / 1,500	1,200 / 1,200
5	*Beginning snowfall*	*"morning" / 9 a.m.*	30 / 3 / 2	16 / 0 / 1 / 0 | 8 / 1 / 4 / 0	3,097	1,500 / 1,500 / 1,500	1,200 / 1,200 / 1,200
6	*Beginning snowfall*	*"morning" / 9 a.m.*	45 / 3 / 2	21 / 0 / 2 / 0 | 16 / 1 / 5 / 0	3,939	1,500 / 1,500 / 1,500	1,200 / 1,200 / 1,200
7	*Beginning snowfall*	*"morning" / 9 a.m.*	60 / 3 / 2	24 / 0 / 2 / 0 | 27 / 1 / 6 / 0	4,819	1,500 / 1,500 / 1,500	1,200 / 1,200 / 1,200
8	*Beginning snowfall*	*"morning" / 9 a.m.*	75 / 3 / 2	31 / 1 / 2 / 0 | 33 / 1 / 7 / 0	5,346	1,500 / 1,500 / 1,500	1,200 / 1,200 / 1,200
9	*Continuous winter operations*	*"noon" / 12 p.m.*	30 / 2 / 1	12 / 0 / 6 / 0 | 10 / 0 / 1 / 1	3,600	600 / 1,500	1,200 / 1,200
10	*Continuous winter operations*	*"noon" / 12 p.m.*	45 / 2 / 1	21 / 0 / 9 / 0 | 13 / 0 / 1 / 1	4,622	600 / 1,500	1,200 / 1,200
11	*Continuous winter operations*	*"noon" / 12 p.m.*	60 / 2 / 1	28 / 0 / 9 / 0 | 20 / 0 / 2 / 1	5,726	600 / 1,500	1,200 / 1,200
12	*Continuous winter operations*	*"noon" / 12 p.m.*	75 / 2 / 1	2 / 0 / 10 / 0 | 30 / 0 / 2 / 1	7,689	600 / 1,500	1,200 / 1,200
13	*Continuous winter operations*	*"noon" / 12 p.m.*	30 / 3 / 2	12 / 0 / 5 / 0 | 11 / 1 / 0 / 1	2,970	600 / 1,500 / 2,400	1,200 / 1,200 / 1,200
14	*Continuous winter operations*	*"noon" / 12 p.m.*	45 / 3 / 2	21 / 0 / 7 / 0 | 14 / 1 / 1 / 1	3,872	600 / 1,500 / 2,400	1,200 / 1,200 / 1,200
15	*Continuous winter operations*	*"noon" / 12 p.m.*	60 / 3 / 2	28 / 0 / 12 / 0 | 17 / 1 / 1 / 1	4,647	600 / 1,500 / 2,400	1,200 / 1,200 / 1,200
16	*Continuous winter operations*	*"noon" / 12 p.m.*	75 / 3 / 2	35 / 1 / 12 / 0 | 22 / 1 / 3 / 1	5,459	600 / 1,500 / 2,400	1,200 / 1,200 / 1,200
17	*Ending snowfall*	*"evening" / 9 p.m.*	30 / 2 / 1	9 / 0 / 2 / 0 | 18 / 1 / 0 / 0	3,489	1,500 / -	1,200 / -
18	*Ending snowfall*	*"evening" / 9 p.m.*	45 / 2 / 1	16 / 0 / 3 / 1 | 22 / 2 / 0 / 1	4,645	1,500 / -	1,200 / -

(*continued on next page*)

Table 6 (*continued*)

Instance	Scenario	Data set	# aircraft / # runways / # snow removal groups	# departing aircraft of class large / Boeing 757 / heavy / super | # landing aircraft of class large / Boeing 757 / heavy / super	Planning horizon $\max_{a\in\mathcal{A}}\{L_a\}$ (in seconds)	Unsafe times U_1 / U_2 / U_3 (in seconds)	Required snow removal times P_1 / P_2 / P_3 (in seconds)
19	*Ending snowfall*	*"evening" / 9 p.m.*	60 / 2 / 1	23 / 0 / 5 / 1 | 28 / 2 / 0 / 1	5,962	1,500 / -	1,200 / -
20	*Ending snowfall*	*"evening" / 9 p.m.*	75 / 2 / 1	34 / 0 / 5 / 1 | 32 / 2 / 0 / 1	7,291	1,500 / -	1,200 / -
21	*Ending snowfall*	*"evening" / 9 p.m.*	30 / 3 / 2	10 / 0 / 2 / 0 | 18 / 0 / 0 / 0	3,058	1,500 / - / 600	1,200 / - / 1,200
22	*Ending snowfall*	*"evening" / 9 p.m.*	45 / 3 / 2	15 / 0 / 2 / 0 | 27 / 1 / 0 / 0	3,841	1,500 / - / 600	1,200 / - / 1,200
23	*Ending snowfall*	*"evening" / 9 p.m.*	60 / 3 / 2	21 / 0 / 3 / 1 | 32 / 2 / 0 / 1	4,683	1,500 / - / 600	1,200 / - / 1,200
24	*Ending snowfall*	*"evening" / 9 p.m.*	75 / 3 / 2	29 / 0 / 5 / 1 | 37 / 2 / 0 / 1	5,635	1,500 / - / 600	1,200 / - / 1,200

Table 7
Characteristics and parameters for large-scale instances.

Instance	Data set	# aircraft / # runways / # snow removal groups	# (landing) aircraft of class small / lower medium / upper medium / heavy	Planning horizon $\max_{a\in\mathcal{A}}\{L_a\}$ (in seconds)	Unsafe times U_1 / U_2 / U_3 / U_4 / U_5 (in seconds)	Required snow removal times P_1 / P_2 / P_3 / P_4 / P_5 (in seconds)
25	*airland9*	100 / 3 / 2	6 / 18 / 30 / 46	14,123	1,200 / 3,600 / 2,400	1,200 / 1,200 / 1,200
26	*airland9*	100 / 4 / 2	6 / 18 / 30 / 46	14,123	1,200 / 3,600 / 2,400 / 4,800	1,200 / 1,200 / 1,200 / 1,200
27	*airland10*	150 / 3 / 2	14 / 28 / 40 / 68	20,815	1,200 / 3,600 / 2,400	1,200 / 1,200 / 1,200
28	*airland10*	150 / 4 / 2	14 / 28 / 40 / 68	20,815	1,200 / 3,600 / 2,400 / 4,800	1,200 / 1,200 / 1,200 / 1,200
29	*airland10*	150 / 5 / 3	14 / 28 / 40 / 68	20,815	1,200 / 3,600 / 2,400 / 4,800 / 6,000	1,200 / 1,200 / 1,200 / 1,200 / 1,200
30	*airland11*	200 / 3 / 2	22 / 34 / 65 / 79	25,799	1,200 / 3,600 / 2,400	1,200 / 1,200 / 1,200
31	*airland11*	200 / 4 / 2	22 / 34 / 65 / 79	25,799	1,200 / 3,600 / 2,400 / 4,800	1,200 / 1,200 / 1,200 / 1,200
32	*airland11*	200 / 5 / 3	22 / 34 / 65 / 79	25,799	1,200 / 3,600 / 2,400 / 4,800 / 6,000	1,200 / 1,200 / 1,200 / 1,200 / 1,200
33	*airland12*	250 / 3 / 2	23 / 54 / 77 / 96	30,629	1,200 / 3,600 / 2,400	1,200 / 1,200 / 1,200
34	*airland12*	250 / 4 / 2	23 / 54 / 77 / 96	30,629	1,200 / 3,600 / 2,400 / 4,800	1,200 / 1,200 / 1,200 / 1,200
35	*airland12*	250 / 5 / 3	23 / 54 / 77 / 96	30,629	1,200 / 3,600 / 2,400 / 4,800 / 6,000	1,200 / 1,200 / 1,200 / 1,200 / 1,200
36	*airland13*	500 / 3 / 2	52 / 94 / 174 / 180	56,383	1,200 / 3,600 / 2,400	1,200 / 1,200 / 1,200
37	*airland13*	500 / 4 / 2	52 / 94 / 174 / 180	56,383	1,200 / 3,600 / 2,400 / 4,800	1,200 / 1,200 / 1,200 / 1,200
38	*airland13*	500 / 5 / 3	52 / 94 / 174 / 180	56,383	1,200 / 3,600 / 2,400 / 4,800 / 6,000	1,200 / 1,200 / 1,200 / 1,200 / 1,200

Table 8
Transfer times between runways (in seconds).

Preceding runway	Succeeding runway				
	Runway 1	Runway 2	Runway 3	Runway 4	Runway 5
Runway 1	-	600	1,200	900	1,200
Runway 2	900	-	1,200	1,200	600
Runway 3	1,500	1,500	-	600	900
Runway 4	1,500	900	900	-	1,500
Runway 5	600	1,200	900	1,200	-

CRediT authorship contribution statement

Maximilian Pohl: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review & editing, Visualization. **Rainer Kolisch:** Conceptualization, Methodology, Investigation, Writing - review & editing, Supervision. **Maximilian Schiffer:** Formal analysis, Methodology, Investigation, Writing - review & editing, Supervision.

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